



THE UNIVERSITY *of* EDINBURGH
Edinburgh Medical School

Tutorial 5

Data Types and Structures in Python and R

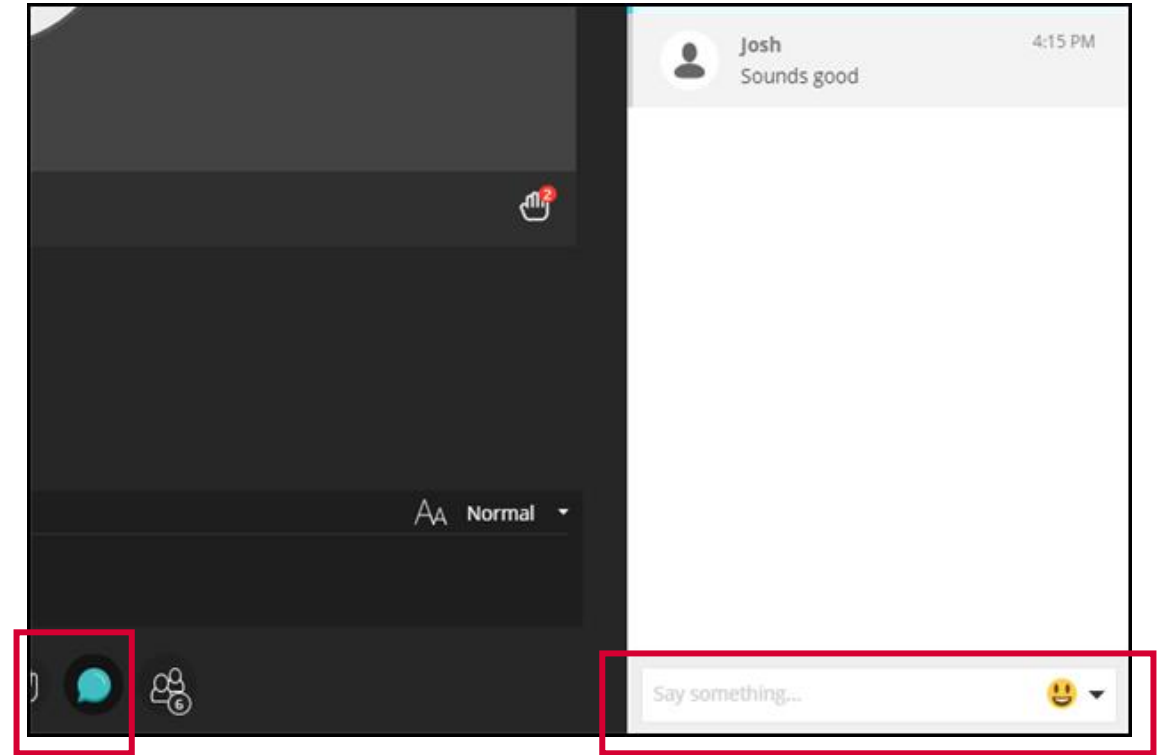
Audio check

Can you hear the presenter talking?

Please type **yes** or **no** in the “Text chat area”

If you can't hear:

- Check your Audio/Visual settings in the Collaborate Panel
- Try signing out and signing back into the session
- Type into the chat box and a moderator will try to assist you



Recording

This session will now be recorded. Any further information that you provide during a session is optional and in doing so you give us consent to process this information.

These sessions will be stored by the University of Edinburgh for one year and published for 30 days after the event. Schools or Services may use the recordings for up to a year on relevant websites.

By taking part in a session, you give us your consent to process any information you provide during it.

A black rectangular button with a white camera icon on the left and the text "Start Recording" in white on the right.

 Start Recording

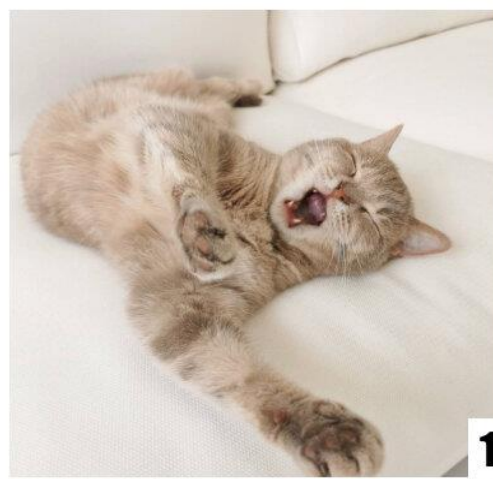


THE UNIVERSITY *of* EDINBURGH
Edinburgh Medical School

Tutorial 5

Data Types and Structures in Python and R

How are you feeling today on this range of expressive animals?



1



2



3



4



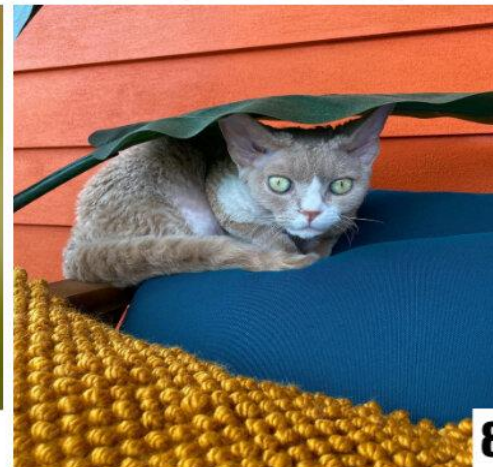
5



6



7



8



9

Agenda

- Do not forget 3 stars, a wish, and a step mini-diaries ★★ ★ 🚀 🦶
- Assessment overview + Q&A – due 29 June 2026 2pm BST
-  Error of the week
-  More on data frames notebooks
-  Any questions
-  [Course feedback survey](#)

Course Assessment



Assessment

This area includes information about the assessments for this course and submission dropboxes. (Click to expand)

- **Assessment Overview and Feedback Information**

Assessment details, important dates, where to find feedback, and other assessment information.

- **Assessment Criteria and Submission Details**

- **Assignment Submission Portal**

Submission portal (Turnitin) for the final assignment, which should be submitted as a ZIP file with all of the necessary documents. PLEASE READ THE ASSIGNMENT SPECIFICATIONS & README CAREFULLY. Submission Due Monday the 29th of June, 2:00pm (BST). Make sure you have put your exam number in the author line and file name of all submissions.

- **Questions about Assessment**

If you have any questions about the assignment, please post them here. FAQ Is my exam number B123456 or BXXXXXX? No! These are placeholders that are obviously incorrect. You must change this to be your exam number both in the file name and in the author lines in both the .ipynb and .RMD files and the overall .zip file. You will be deducted points if you do not include your exam number correctly in your submission (this means do not include your name, your student number, or the placeholder exam number). Can I solve both tasks in the same language? No. There are 2 coding tasks and you must chose which language to use to solve them. But you cannot use the same programming language for both tasks. So if you choose to solve Task 1 in ...

- Assessment folder now on course GitHub repository
- See Assessment Criteria and Submission Details page on Learn
- Due 29 June 2026 14:00 BST (UCT+1)
- Submit via **Turnitin**
- Submit .zip file with:
 1. Submission Declaration
 2. .Rproj file
 3. .RMD file – R Task & Report
 4. .ipynb file – Python Task & Report

Assessment outline

- Coding tasks and accompanying report
- 2 tasks to complete – 1 in Python and 1 in R
 - YOU decide which language to use for each task
 - Task 1 in Python & Task 2 in R **OR** Task 1 in R and Task 2 in Python
- Report accompanying each task
 - **Word Limit:** ~550 words for each task, a total of 1100 words maximum across both tasks
 - What is the impact of your decision of programming language on your approach to solving the task?
 - What would you have done differently if you chose the other programming language to solve this task? How would your approach change? What other decisions would you have made?
 - What data types did you select to use in solving the task and why did you decide against other data types in the programming language?
 - What data structure(s) did you select to use in solving the task and why did you decide against other data structure(s) in the programming language?

Submission requirements

- 1st step: rename the files and author lines with your exam number

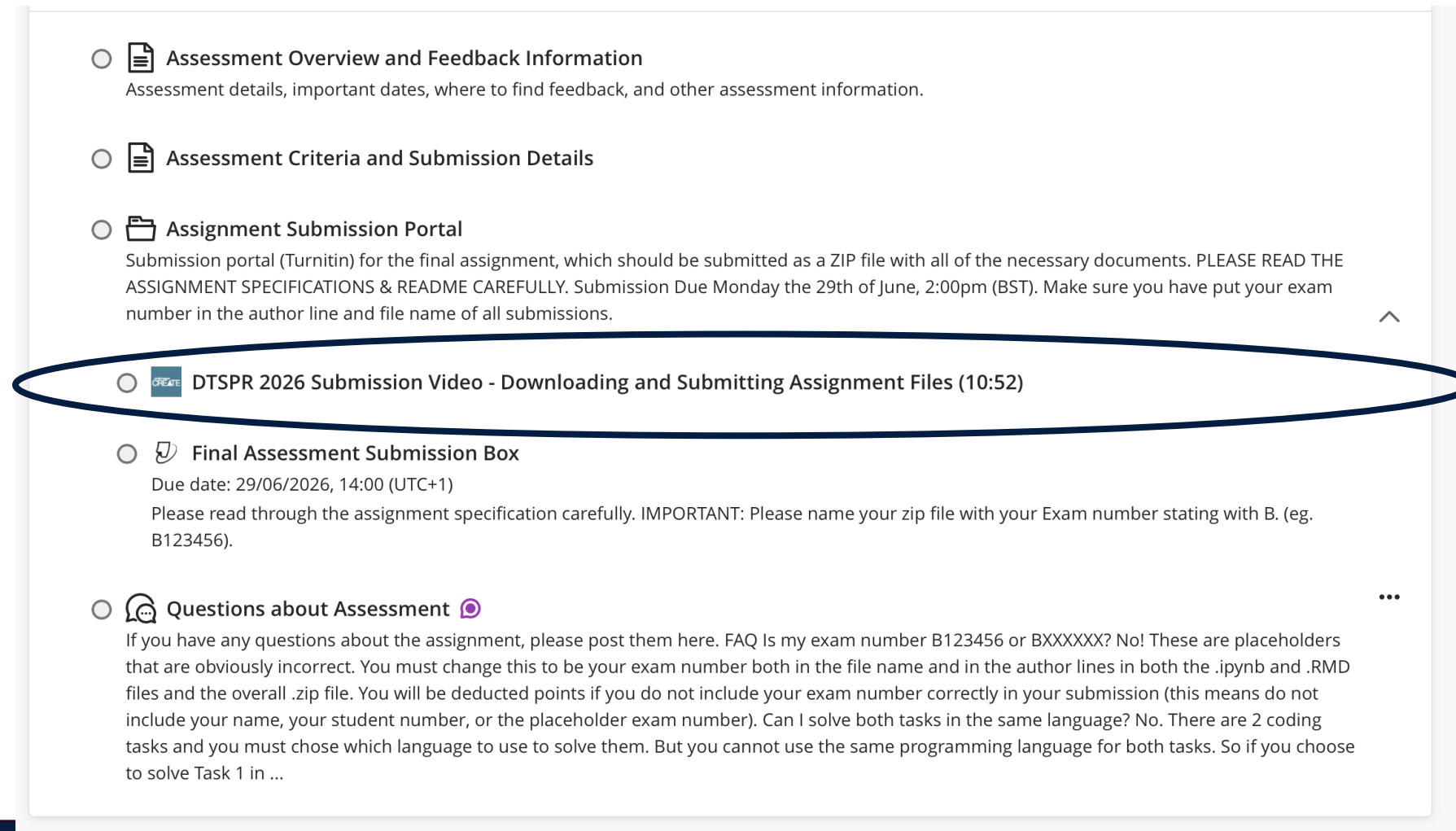
Your exam number is listed at **MyEd > Accounts > Progression > My Student Record**.

- ! Points will be deducted on submissions that in do not include exam numbers and/or include names or student numbers

The screenshot displays the 'My Student Record' page in the MyEd system. It features two main sections: 'Current Addresses' and 'Emergency Contacts'. The 'Current Addresses' section is divided into 'Home' and 'Semester' sub-sections. The 'Home' section has a text input field and an 'Add new Address' button. The 'Semester' section has a text input field with a note: 'If you have a place in University accommodation you do not need to add a semester address. It will appear here automatically in September.' and an 'Add new Address' button. The 'Emergency Contacts' section has a 'First' text input field, an 'Edit Contact One' button, and an 'Add' button. At the bottom, there is a section for 'Personal Information' with fields for 'Full Name', 'Known as', 'Date of Birth', 'Exam Number', 'Country of Birth', and 'Nationality'. The 'Exam Number' field is highlighted with a red circle. A note at the bottom states: 'If you need to update any information below that cannot be edited from this screen please refer to the [Change Student Records webpage](#).'

Submission requirements

- See video for more details on downloading and submitting files



The screenshot shows a list of submission-related items in a portal. A hand-drawn blue oval highlights the item 'DTSPR 2026 Submission Video - Downloading and Submitting Assignment Files (10:52)'. The items are as follows:

-  **Assessment Overview and Feedback Information**
Assessment details, important dates, where to find feedback, and other assessment information.
-  **Assessment Criteria and Submission Details**
-  **Assignment Submission Portal**
Submission portal (Turnitin) for the final assignment, which should be submitted as a ZIP file with all of the necessary documents. PLEASE READ THE ASSIGNMENT SPECIFICATIONS & README CAREFULLY. Submission Due Monday the 29th of June, 2:00pm (BST). Make sure you have put your exam number in the author line and file name of all submissions.
-  **DTSPR 2026 Submission Video - Downloading and Submitting Assignment Files (10:52)**
-  **Final Assessment Submission Box**
Due date: 29/06/2026, 14:00 (UTC+1)
Please read through the assignment specification carefully. IMPORTANT: Please name your zip file with your Exam number stating with B. (eg. B123456).
-  **Questions about Assessment** 

Marking criteria

By submitting the assessment, students demonstrate LO1.

Criteria	Description	Weighting
Code quality [LO2]	• Code is clean, meaningful, and well structured, integrating good coding practice. • An understanding of Python and R and related libraries/modules/packages taught on the course is clearly demonstrated. • At this level, we do not expect to see highly refactored or optimized code.	35%
Problem solving and approach [LO3]	• Evidence of analytical thinking and problem decomposition in completing the coding tasks, integrating what has been taught in the course	30%
Communication [LO4]	• Task reports are well reasoned and explained. • Decisions around programming languages are clear and limitations and strengths are clearly and accurately communicated. • Discussion is presented in clear academic writing showing critical analysis and understanding of the process of completing the tasks, how the programming languages are similar and different, and the larger context of data types and structures.	35%



Any questions?

!?! Error of the week !?!

Common errors when pivoting data frames

Course Feedback Survey

- Takes 5 minutes or less!
- Each response is read individually by the course team
- Feedback used to create adjustments to the course for next year
 - This is how the Data Dialogues series came about, for example!
- The only official way we know if we are doing a good job, so please let us know!
- Tell us what brought you joy and what you wish was different

Give a gift to Brittany and the course team (survey link) here 

Working with data structures notebooks

- 3 files for this tutorial
 1. Tutorial05-Python.ipynb
 2. Tutorial05-R.RMD
 3. Map-HealthBoards.png
- Solutions notebooks released after the Friday tutorial
 - Really do try the exercises without looking at the solutions – trial and error is the best way to learn!